Exact power tails and moment growth for a contractive nonlinear SGD model

Yitzchak Shmalo

Einstein Institute of Mathematics, The Hebrew University of Jerusalem
Givat Ram, Jerusalem, Israel

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Abstract

For the smooth nonlinear activation $\sigma(z) = z + 10^{-3}\tanh z$, we study ordinary constant-step, batch-one squared-loss SGD with Gaussian inputs, independent Gaussian label noise, and regression teacher $\theta \in [-1, 1]$. For every fixed $\eta \in [0.99, 1.01]$, the actual random update maps satisfy $\mathbb{E}\,\mathrm{Lip}(\Psi_\theta) < 0.99010707 < 1$. There is a unique stationary law, geometric convergence in Wasserstein distance of order one, and a finite strictly positive exact absolute-tail constant. The tail exponent is the unique $\alpha \in (1, 2)$ satisfying $\mathbb{E}|1 - \eta A^2|^\alpha = 1$; it does not depend on the teacher. We verify every input to the scalar implicit renewal theorem and give the full forward-excursion proof of strict positivity. For deterministic initialization, we also prove exact linear growth of the critical moment and positive finite limits for appropriately normalized supercritical moments. The result is a fully verified nonlinear application of established implicit renewal theory; it is not a general theorem for deep neural networks.

1 Introduction, model, and main theorem

Multiplicative fluctuations can give stochastic gradient descent a heavy-tailed stationary distribution even when the driving data have all moments. Gürbüzbalaban, Şimşekli and Zhu [2] develop this mechanism for SGD and formulate a general nonlinear criterion using the theory of iterated random maps. Our contribution is an explicit smooth nonlinear scalar model for which average contraction, exact tail asymptotics, and strict positivity of the tail constant are all verified on a nonempty parameter rectangle.

The activation is the identity plus a bounded smooth perturbation; the activation itself is unbounded. The model has persistent Gaussian label noise and a possibly nonzero regression teacher. A forward excursion argument proves that the exact tail constant cannot vanish. We also derive critical and supercritical finite-time moment asymptotics. These are consequences for this model, not claims of a new general implicit renewal theorem or of priority for the heavy-tail mechanism.

Fix, throughout a trajectory,

$$\delta = \frac{1}{1000}, \qquad \sigma(z) = z + \delta\tanh z, \qquad \theta \in [-1, 1], \qquad \eta \in \left[\frac{99}{100}, \frac{101}{100}\right], \qquad b = 1. \tag{1.1}$$

Let $(A_k, E_k)_{k\ge1}$ be independent and identically distributed pairs, each pair consisting of two independent standard normal variables. Set $Y_k = \sigma(A_k\theta) + E_k$. For a fresh pair (A, E) and its label $Y = \sigma(A\theta) + E$, define

$$f_\theta(x; A, E) = \frac{1}{2}\big(\sigma(Ax) - \sigma(A\theta) - E\big)^2, \tag{1.2}$$

$$\Psi_\theta(x) = x - \eta A\sigma'(Ax)(\sigma(Ax) - Y), \qquad X_{k+1} = \Psi_{\theta,k+1}(X_k). \tag{1.3}$$

This is ordinary, unclipped, constant-step SGD on the sample loss (1.2). For $\theta \neq 0$, labels and inputs are dependent.

Theorem 1.1 (Explicit nonlinear model for T228)**.** *For every fixed pair (η, θ) in (1.1), the following conclusions hold.*

(i) *The smooth sample maps are nonaffine almost surely, and*

$$\mathbb{E}\,\mathrm{Lip}(\Psi_\theta) < q := \frac{99010707}{100000000} = 0.99010707 < 1. \tag{1.4}$$

They therefore contract on average, also in the logarithmic sense.

(ii) *There is exactly one invariant probability law $\pi_{\eta,\theta}$, including among laws without an assumed first moment. It has a finite first moment, has support $\mathbb{R}$, and for every deterministic $x \in \mathbb{R}$,*

$$W_1\big(\mathrm{Law}(X_n^x), \pi_{\eta,\theta}\big) \le q^n W_1\big(\delta_x, \pi_{\eta,\theta}\big). \tag{1.5}$$

Here δ_x is the unit point mass at x.

(iii) *There is exactly one positive solution $\alpha = \alpha(\eta)$ of*

$$h_\eta(\alpha) := \mathbb{E}|1 - \eta A^2|^\alpha = 1, \qquad 1 < \alpha < 2. \tag{1.6}$$

The law of $\log|1 - \eta A^2|$ is nonarithmetic. The exponent is independent of θ.

(iv) *A finite strictly positive constant $C_{\eta,\theta}$ satisfies*

$$\lim_{t\to\infty} t^\alpha \pi_{\eta,\theta}\big(\{x : |x| > t\}\big) = C_{\eta,\theta} \in (0, \infty). \tag{1.7}$$

The same limit and constant hold with $|x - \theta|$ in place of $|x|$. For every $p > 0$, both the absolute pth moment and the estimation-error pth moment are finite exactly when $p < \alpha$. Their second moments are infinite.

If $X \sim \pi_{\eta,\theta}$ is independent of the fresh step (A, E), then

$$C_{\eta,\theta} = \frac{\mathbb{E}\big[|\Psi_\theta(X)|^\alpha - |(1 - \eta A^2)X|^\alpha\big]}{\alpha\,\mathbb{E}[|1 - \eta A^2|^\alpha \log|1 - \eta A^2|]}. \tag{1.8}$$

The numerator is an absolutely integrable coupled difference*, not a subtraction of two separately finite expectations.*

Remark 1.2 (Symmetry and uniformity). When $\theta = 0$, the stationary law is symmetric and each one-sided tail has constant $C_{\eta,0}/2$. No one-sided tail limit or equality of one-sided constants is asserted for a fixed nonzero teacher. Indeed, its stationary law is not symmetric about zero; see section 10. The contraction bound is uniform on the stated rectangle. No uniform positive lower bound for $C_{\eta,\theta}$ is claimed.

The proof of theorem 1.1 occupies sections 2 to 8. The nonzero-signal example $\theta = \eta = 1$ is made explicit in section 9.

2 Bounds on the actual nonlinear update

Lemma 2.1 (Global derivative and uniform remainder)**.** *Put $M = 1 - \eta A^2$ and $\xi = |M|$. Define*

$$L_\theta = \xi + \eta A^2(4\delta + 3\delta^2 + 2\delta|Y|), \tag{2.1}$$

$$B_\theta = \eta|A|\big[\delta + (1 + \delta)(\delta + |Y|)\big]. \tag{2.2}$$

Then $\operatorname{Lip}(\Psi_\theta) \le L_\theta$ *and*

$$\Psi_\theta(x) = Mx + R_\theta(x), \qquad \sup_{x\in\mathbb{R}} |R_\theta(x)| \le B_\theta. \tag{2.3}$$

Both L_θ *and* B_θ *have moments of every positive order. The map* Ψ_θ *is nonaffine almost surely.*

Proof. Write $z = Ax$, $t = \tanh z$, and $u = \operatorname{sech}^2 z$. Differentiating the *actual* update yields

$$\begin{aligned} \sigma'(z) &= 1 + \delta u, & \sigma''(z) &= -2\delta u t, \\ \Psi'_\theta(x) &= 1 - \eta A^2 H(z,Y), & H(z,Y) &= (1+\delta u)^2 - 2\delta u t(z + \delta t - Y). \end{aligned} \tag{2.4}$$

We have $|t| \le 1$, $0 \le u \le 1$, and $|z|u \le 1$. For the last bound, $\cosh^2 z = 1+\sinh^2 z \ge 1+z^2 \ge |z|$. Expansion gives

$$\begin{aligned} |H - 1| &\le 2\delta u + \delta^2 u^2 + 2\delta |z| u |t| + 2\delta^2 u t^2 + 2\delta u |t| |Y| \\ &\le 4\delta + 3\delta^2 + 2\delta |Y|. \end{aligned} \tag{2.5}$$

The mean value theorem on $\mathbb{R}$ proves the global Lipschitz bound (2.1). It bounds the actual nonlinear map, not merely its asymptotic multiplier.

For the remainder, use the exact identity

$$\sigma'(z)(\sigma(z) - Y) = z + \delta z u + (1+\delta u)(\delta t - Y). \tag{2.6}$$

Multiplication by ηA and the preceding bounds give (2.3) with (2.2).

The label envelope keeps the input-label dependence:

$$|Y| \le (1+\delta)|\theta||A| + |E|. \tag{2.7}$$

In particular, throughout the parameter rectangle,

$$B_\theta \le \frac{101}{100}|A|\big[\delta + (1+\delta)(\delta + (1+\delta)|A| + |E|)\big]. \tag{2.8}$$

The envelopes for B_θ, L_θ, and Y are polynomials in $|A|$ and $|E|$, hence have all positive moments. Neither B_θ nor Y is required to be independent of M.

At zero, the derivative is $1 - \eta A^2(1+\delta)^2$. For $A \ne 0$, the terms involving u or zu vanish as $|x| \to \infty$, and $\Psi'_\theta(x) \to 1 - \eta A^2$. These values differ for every $A \ne 0$, an event of probability one. Thus the map is nonaffine almost surely. □

3 Exact contraction arithmetic and the exponent

Lemma 3.1 (Uniform average contraction)**.** *The constant* q *in* (1.4) *bounds* $\mathbb{E}L_\theta$ *strictly. Also* $\mathbb{E}\log L_\theta$ *exists as a finite number and is negative; in particular,* $\mathbb{E}\log \operatorname{Lip}(\Psi_\theta) < 0$ *in the extended sense.*

Proof. Let φ be the standard normal density. Splitting at $|A| = 1$, using $\mathbb{E}(1 - A^2) = 0$ and $(a\varphi(a))' = (1 - a^2)\varphi(a)$, gives

$$\mathbb{E}|1 - A^2| = 4\varphi(1) = \frac{4}{\sqrt{2\pi e}} < \frac{97}{100}. \tag{3.1}$$

The strict inequality needs no numerical quadrature. The elementary bounds $\pi > 157/50$ and $e > 1359/500$ imply

$$\left(\frac{97}{100}\right)^2 2\frac{157}{50}\frac{1359}{500} - 16 = \frac{7532467}{125000000} > 0. \tag{3.2}$$

Since $|1-\eta A^2| \le |1-A^2| + |\eta-1|A^2$, it follows that

$$\mathbb{E}\xi < \frac{98}{100}. \tag{3.3}$$

The mixed label moment must retain the dependence between A and Y. By (2.7),

$$\begin{aligned}\mathbb{E}[A^2|Y|] &\le (1+\delta)|\theta|\mathbb{E}|A|^3 + \mathbb{E}[A^2|E|] \\ &= (1+\delta)|\theta|\,2\sqrt{\frac{2}{\pi}} + \sqrt{\frac{2}{\pi}} \le (3+2\delta)\sqrt{\frac{2}{\pi}} < 3+2\delta = \frac{1501}{500}.\end{aligned} \tag{3.4}$$

Only the second summand uses independence, namely that of A and E. There is no factorization of $\mathbb{E}[A^2|Y|]$. Thus

$$\begin{aligned}\mathbb{E}\,\mathrm{Lip}(\Psi_\theta) \le \mathbb{E}L_\theta &< \frac{98}{100} + \frac{101}{100}\left[\frac{4}{1000} + \frac{3}{10^6} + \frac{2}{1000}\frac{1501}{500}\right] \\ &= \frac{98}{100} + \frac{101}{100}\frac{10007}{10^6} = \frac{99010707}{10^8} = q < 1.\end{aligned} \tag{3.5}$$

The exact margin is $1-q = 989293/10^8 = 0.00989293$.

Jensen's inequality gives $\mathbb{E}\log\mathrm{Lip}(\Psi_\theta) \le \log\mathbb{E}\,\mathrm{Lip}(\Psi_\theta) < \log q < 0$, allowing a negative infinite expectation. For the envelope itself, set $c_0 = 4\delta + 3\delta^2 < 1$. With $v = \eta A^2 \ge 0$, $L_\theta \ge |1-v| + c_0 v \ge c_0$, whereas $\log^+ L_\theta \le L_\theta$. Hence $\mathbb{E}|\log L_\theta| < \infty$ and $\mathbb{E}\log L_\theta < \log q < 0$. □

Lemma 3.2 (Lyapunov root and logarithmic coefficient)**.** *There is a unique positive root $\alpha \in (1,2)$ of $h_\eta(p) = \mathbb{E}\xi^p = 1$. For $0 < p < \alpha$, $h_\eta(p) < 1$. Moreover*

$$m_\alpha := \mathbb{E}[\xi^\alpha \log\xi] \in (0,\infty), \tag{3.6}$$

and $\log\xi$ is nonarithmetic.

Proof. The distribution of $M = 1-\eta A^2$ is absolutely continuous. Its zeros as a function of A are $\pm\eta^{-1/2}$, with nonzero derivatives there. Local logarithmic integrability near these roots and Gaussian decay at infinity give

$$\mathbb{P}(\xi = 0) = 0, \qquad \mathbb{E}|\log\xi| < \infty, \qquad \mathbb{E}[\xi^p|\log\xi|] < \infty \quad (p > 0). \tag{3.7}$$

The law of $\log\xi$ is absolutely continuous and thus nonarithmetic. Also $\mathbb{E}\log\xi \le \log\mathbb{E}\xi < 0$.

The function h_η is finite for all $p \ge 0$, and its logarithm is strictly convex because $\log\xi$ is nonconstant. We have

$$\begin{aligned}&h_\eta(0) = 1, && h_\eta(1) < 0.98 < 1, \\ &h_\eta(2) = 1 - 2\eta + 3\eta^2 \ge 1 - 2\frac{99}{100} + 3\left(\frac{99}{100}\right)^2 = \frac{19603}{10000} > 1.\end{aligned} \tag{3.8}$$

The quadratic is increasing on the permitted interval. Since $\mathbb{P}(\xi > 1) > 0$, $h_\eta(p) \to \infty$ as $p \to \infty$. Strict log-convexity therefore yields exactly one positive root, lying between one and two, and $h_\eta(p) < 1$ for $0 < p < \alpha$. The integrability in (3.7) makes m_α finite. The derivative of $\log h_\eta$ at α is at least its strictly positive secant slope from 1 to α, so $m_\alpha > 0$. All of these quantities depend on η and not on θ. □

Equivalently, the equation specifying the index is

$$\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} |1-\eta a^2|^\alpha e^{-a^2/2}\,\mathrm{d}a = 1. \tag{3.9}$$

For a general batch b, the corresponding scalar asymptotic multiplier is $1-(\eta/b)\chi_b^2$. The present theorem fixes $b = 1$; the interval in (1.1) already supplies a nonempty regime.

4 Invariant law, uniqueness, and lower moments

Proposition 4.1 (Invariant law and convergence)**.** *There is a unique invariant probability $\pi = \pi_{\eta,\theta}$ on $\mathbb{R}$. It has a finite first moment, satisfies (1.5), and*

$$\mathbb{E}_\pi |X|^p < \infty \qquad (0 < p < \alpha). \tag{4.1}$$

Proof. At zero,

$$\Psi_\theta(0) = \eta(1+\delta)AY, \qquad \mathbb{E}|\Psi_\theta(0)| < \infty. \tag{4.2}$$

The Markov operator consequently maps probability laws with finite first moment into themselves. Coupling two inputs by the same independent fresh step gives

$$W_1(\mu P_\theta, \nu P_\theta) \le (\mathbb{E}L_\theta) W_1(\mu,\nu) \le q W_1(\mu,\nu). \tag{4.3}$$

Completeness of the space of such laws for W_1 and the contraction mapping theorem provide an invariant law in that space and the geometric bound (1.5).

To prove uniqueness without a hidden moment restriction, start a second chain from any finite random value Z, independent of its future steps, and couple it to the chain from zero using the same steps. Then

$$|X_n^Z - X_n^0| \le |Z| \prod_{j=1}^n L_{\theta,j} \longrightarrow 0 \quad \text{almost surely.} \tag{4.4}$$

Indeed, the product has expectation $(\mathbb{E}L_\theta)^n \le q^n$; Markov's inequality followed by Borel–Cantelli proves that the product tends to zero. If Z has any invariant law, bounded Lipschitz test functions and dominated convergence identify that law with π. No moment of Z is used.

For $0 < p \le 1$ with $p < \alpha$, (2.3) and subadditivity give

$$\mathbb{E}|\Psi_\theta(x)|^p \le h_\eta(p)|x|^p + \mathbb{E}B_\theta^p. \tag{4.5}$$

For $1 < p < \alpha$, use $(a+b)^p \le (1+\varepsilon)a^p + C_{p,\varepsilon} b^p$ and choose $\varepsilon > 0$ so that $r_p = (1+\varepsilon)h_\eta(p) < 1$. In both cases this yields

$$\mathbb{E}|\Psi_\theta(x)|^p \le r_p|x|^p + d_p, \qquad r_p < 1, \quad d_p < \infty. \tag{4.6}$$

The chain started from zero has uniformly bounded pth moments. Its weak convergence to π and the lower semicontinuity of $|x|^p$ imply (4.1). □

5 Explicit compact minorization and full support

Lemma 5.1 (Compact-to-compact minorization)**.** *For every finite $R, T > 0$ there is an explicit $\varepsilon_{R,T} > 0$, uniform over the parameter rectangle, such that for every Borel F,*

$$P_\theta(x, F) \ge \varepsilon_{R,T} \operatorname{Leb}(F \cap [-T, T]) \qquad (|x| \le R). \tag{5.1}$$

Consequently $\operatorname{supp} \pi_{\eta,\theta} = \mathbb{R}$.

Proof. Conditionally on $A = a$, the fresh E remains standard normal and

$$\begin{aligned} \Psi_\theta(x) &= \mu_\theta(x,a) + \eta a \sigma'(ax) E, \\ \mu_\theta(x,a) &= x - \eta a \sigma'(ax)[\sigma(ax) - \sigma(a\theta)]. \end{aligned} \tag{5.2}$$

For $a \in [1,2]$ and $|x| \le R$, the inequalities $\sigma' \le 1+\delta$ and $|\sigma(v)| \le (1+\delta)|v|$ imply

$$|\mu_\theta(x,a)| \le D_R := R + 4\frac{101}{100}(1+\delta)^2(R+1). \tag{5.3}$$

The conditional standard deviation belongs to

$$[s_{\min}, s_{\max}] := \left[\frac{99}{100},\ 2\frac{101}{100}(1+\delta)\right]. \tag{5.4}$$

At each $|y| \le T$ its density is therefore at least

$$d_{R,T} := \frac{1}{s_{\max}\sqrt{2\pi}} \exp\left[-\frac{(T+D_R)^2}{2s_{\min}^2}\right] > 0. \tag{5.5}$$

Integrate over $a \in [1,2]$ and set $\varepsilon_{R,T} = \mathbb{P}(1 \le A \le 2) d_{R,T} > 0$ to obtain (5.1). In particular, every nonempty open interval has positive transition probability from every point. Integrating this strictly positive transition probability against π proves that every such interval has positive stationary mass. □

6 Exact absolute-tail limit and the constant formula

The external tail-limit input is the $N = 1$ specialization of Jelenković–Olvera-Cravioto [1, Theorem 3.1, printed p. 5]. This is a discrepancy theorem for a nonnegative variable independent of its coefficient. Its use does not require the nonlinear update itself to be an affine fixed-point recursion.

Lemma 6.1 (The scalar renewal input)**.** *Let $R_0 \ge 0$ be independent of $\xi > 0$. Suppose $\log \xi$ is nonarithmetic, $\mathbb{E}\xi^\alpha = 1$, $0 < \mathbb{E}[\xi^\alpha \log \xi] < \infty$, $\mathbb{E}\xi^\gamma < \infty$ for some $0 \le \gamma < \alpha$, and $\mathbb{E}R_0^p < \infty$ for every $0 < p < \alpha$. If*

$$\int_0^\infty t^{\alpha-1} \left|\mathbb{P}(R_0 > t) - \mathbb{P}(\xi R_0 > t)\right| \, \mathrm{d}t < \infty, \tag{6.1}$$

then $\lim_{t\to\infty} t^\alpha \mathbb{P}(R_0 > t) = C \in [0,\infty)$, where

$$C = \frac{1}{m_\alpha} \int_0^\infty t^{\alpha-1} [\mathbb{P}(R_0 > t) - \mathbb{P}(\xi R_0 > t)] \, \mathrm{d}t. \tag{6.2}$$

Proof. Apply [1, Theorem 3.1] with $N = 1$ and $C_1 = \xi$. The stated conditions are its scalar hypotheses. The conclusion is expressed as a scaled limit, also covering the possibility $C = 0$. □

Proposition 6.2 (Existence of the finite tail limit)**.** *The limit in (1.7) exists in $[0,\infty)$ and has the formula (1.8). Its numerator is absolutely integrable.*

Proof. Choose $X \sim \pi$ independently of the fresh (A,E) and take $R_0 = |X|$. Then R_0 is independent of ξ. Lemma 3.2 provides the multiplier equation, logarithmic nonarithmeticity, and $0 < m_\alpha < \infty$. Taking $\gamma = 0$ supplies the auxiliary coefficient moment. Proposition 4.1 supplies every stationary moment below α. Thus only (6.1) remains.

Let $U = |\Psi_\theta(X)|$ and $W = |MX| = \xi|X|$. Stationarity gives U the law of R_0, and (2.3) gives $|U - W| \le B_\theta$. Tonelli's theorem and the indicator-tail integral yield

$$\begin{aligned}\int_0^\infty t^{\alpha-1} &\left|\mathbb{P}(R_0 > t) - \mathbb{P}(\xi R_0 > t)\right| \, \mathrm{d}t \\ &\le \mathbb{E}\int_0^\infty t^{\alpha-1} |\mathbf{1}_{\{U>t\}} - \mathbf{1}_{\{W>t\}}| \, \mathrm{d}t = \frac{1}{\alpha}\mathbb{E}|U^\alpha - W^\alpha|.\end{aligned} \tag{6.3}$$

Because $1 < \alpha < 2$, the mean value theorem and subadditivity at exponent $\alpha - 1$ imply

$$\begin{aligned} |U^\alpha - W^\alpha| &\le \alpha B_\theta (W + B_\theta)^{\alpha-1} \\ &\le \alpha \left[B_\theta^\alpha + B_\theta \xi^{\alpha-1} |X|^{\alpha-1} \right]. \end{aligned} \tag{6.4}$$

The right side is integrable. The stationary moment of order $\alpha - 1$ is finite; X is independent of the step; and the polynomial Gaussian envelopes give $\mathbb{E}[B_\theta \xi^{\alpha-1}] < \infty$. This is a *joint* moment: independence of B_θ and ξ is neither asserted nor needed.

Lemma 6.1 now gives a finite nonnegative limit. Signed Fubini is legitimate by (6.4), and gives

$$\int_0^\infty t^{\alpha-1} [\mathbb{P}(U > t) - \mathbb{P}(W > t)] \, \mathrm{d}t = \frac{1}{\alpha} \mathbb{E}[U^\alpha - W^\alpha]. \tag{6.5}$$

Substitution into (6.2) proves (1.8). Strict positivity requires the separate argument that follows. □

7 Strict positivity by forward excursions

This section gives the complete positivity mechanism supplied by an earlier project Claude agent, with every bound checked for the nonzero-teacher law. The contribution is credited in Section 12. The steps (A_j, E_j) are iid, so $(\xi_j, B_{\theta,j})$ are iid pairs. Their two components may be dependent. The tilt below acts on the *complete step*; no teacher symmetry or rotation argument is used.

7.1 Radial comparison and the stopped likelihood ratio

Set

$$S_0 = 0, \qquad S_n = \sum_{j=1}^n \log \xi_j, \qquad V_n = \sum_{j=1}^n e^{-S_j} B_{\theta,j}. \tag{7.1}$$

Repeated reverse triangle inequalities in (2.3) give, pathwise,

$$|X_n| \ge e^{S_n} (|X_0| - V_n). \tag{7.2}$$

To see this directly, multiply $|X_j| \ge \xi_j |X_{j-1}| - B_{\theta,j}$ by e^{-S_j} and sum the resulting telescoping inequalities.

Tilt each complete step by ξ^α and call the iid product law $\mathbb{Q}$. It is a probability law because $\mathbb{E}\xi^\alpha = 1$. On the sigma-field $\mathcal{F}_n$ of the first n complete steps, $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = e^{\alpha S_n}$. For any stopping time τ and stopped event $F \in \mathcal{F}_\tau$, summing over the disjoint events $\{\tau = n\}$ proves

$$\mathbb{P}(F, \tau < \infty) = \mathbb{E}_\mathbb{Q} \left[e^{-\alpha S_\tau} \mathbf{1}_{F \cap \{\tau < \infty\}} \right]. \tag{7.3}$$

This identity makes no infinite-path absolute-continuity or uniform- integrability assertion. Under $\mathbb{Q}$, the increment $Z = \log \xi$ satisfies $\mathbb{E}_\mathbb{Q}|Z| < \infty$ and $\mathbb{E}_\mathbb{Q} Z = m_\alpha > 0$. Hence $S_n \to \infty$ almost surely under $\mathbb{Q}$. Fix

$$\beta = \frac{1}{2}, \qquad r := \mathbb{E}_\mathbb{Q} e^{-\beta Z} = h_\eta(\alpha - \beta) < 1. \tag{7.4}$$

The inequality holds because $0 < \alpha - \beta < \alpha$.

7.2 Positive survival, tight overshoots, and finite remainder

Lemma 7.1 (Three tilted-walk estimates)**.** *Under $\mathbb{Q}$, the following hold:*

(a) *The positive-survival probability obeys*

$$p_+ := \mathbb{Q}(S_j > 0 \text{ for every } j \geq 1) \geq p_0 := \mathbb{Q}(Z > 1)(1 - e^{-\beta}) > 0. \tag{7.5}$$

(b) *If $\tau_u = \inf\{n \geq 1 : S_n > u\}$ for $u \geq 0$ and $O_u = S_{\tau_u} - u$, then the overshoots O_u are uniformly tight:*

$$\sup_{u \geq 0} \mathbb{Q}(O_u > z) \leq K_U \left[\mathbb{Q}(Z > z) + \int_z^\infty \mathbb{Q}(Z > v)\, \mathrm{d}v\right] \longrightarrow 0, \qquad z \to \infty, \tag{7.6}$$

where $K_U = e^\beta/(1 - r)$.

(c) *The discounted remainder is finite almost surely, and*

$$\mathbb{E}_{\mathbb{Q}} V_\infty^\beta \leq \frac{\mathbb{E}_{\mathbb{P}}[\xi^{\alpha-\beta} B_\theta^\beta]}{1 - h_\eta(\alpha - \beta)} < \infty, \qquad V_\infty := \sum_{j \geq 1} e^{-S_j} B_{\theta,j}. \tag{7.7}$$

Proof. (a) Survival. By (7.4), $e^{-\beta S_n}$ is a nonnegative supermartingale. Stop it at the first time an independent walk reaches $(-\infty, -1]$, truncated at n. On the event that this time is at most n, its stopped value is at least e^β. The supermartingale inequality and then monotone convergence give

$$\mathbb{Q}\Big(\inf_{n \geq 0} S_n \leq -1\Big) \leq e^{-\beta}. \tag{7.8}$$

A first increment exceeding one followed by a fresh walk that never reaches -1 stays strictly positive. Independence proves (7.5). The Gaussian input is unbounded, and the tilt has strictly positive density where $\xi > 0$, so $\mathbb{Q}(Z > 1) > 0$.

(b) Overshoots. Positive drift makes τ_u finite almost surely under $\mathbb{Q}$. Chernoff's inequality and (7.4) give

$$\sum_{n \geq 0} \mathbb{Q}(S_n \leq 1) \leq \sum_{n \geq 0} e^\beta r^n = \frac{e^\beta}{1 - r} = K_U. \tag{7.9}$$

For $I = (a, a + 1]$, let ν_I be its first entry time, allowing time zero. Conditionally on $S_{\nu_I} \in I$, a later visit to I entails a fresh-walk displacement at most one. The strong Markov property and (7.9) therefore give the translated occupation bound

$$U(I) := \sum_{n \geq 0} \mathbb{Q}(S_n \in I) \leq K_U. \tag{7.10}$$

Partition $(-\infty, u]$ into the intervals $(u - k - 1, u - k]$, $k \geq 0$. Dropping the restriction that no earlier crossing has occurred, and using independence of the next increment from the past, yields for $z \geq 0$

$$\begin{aligned} \mathbb{Q}(O_u > z) &\leq \sum_{n \geq 1} \mathbb{E}_{\mathbb{Q}}\Big[\mathbf{1}_{\{S_{n-1} \leq u\}}\, \mathbb{Q}(Z > u + z - S_{n-1})\Big] \\ &\leq K_U \sum_{k \geq 0} \mathbb{Q}(Z > z + k) \\ &\leq K_U \left[\mathbb{Q}(Z > z) + \int_z^\infty \mathbb{Q}(Z > v)\, \mathrm{d}v\right]. \end{aligned} \tag{7.11}$$

The last expression is independent of u and tends to zero because $\mathbb{E}_{\mathbb{Q}} Z^+ < \infty$.

(c) Discounted remainders. Subadditivity at $\beta = 1/2$, Tonelli's theorem, and independence of a complete step from its past imply

$$\mathbb{E}_{\mathbb{Q}}\left(\sum_{j\geq 1} e^{-S_j} B_{\theta,j}\right)^{\beta} \leq \sum_{j\geq 1} r^{j-1}\mathbb{E}_{\mathbb{Q}}[\xi^{-\beta}B_\theta^\beta] = \frac{\mathbb{E}_{\mathbb{P}}[\xi^{\alpha-\beta}B_\theta^\beta]}{1-h_\eta(\alpha-\beta)} < \infty. \tag{7.12}$$

The numerator is the correct joint moment. It is finite by the polynomial Gaussian envelopes and $\alpha - \beta > 0$; there is no inverse-multiplier singularity. Finite βth moment implies $V_\infty < \infty$ almost surely under $\mathbb{Q}$. □

7.3 A uniform excursion lower bound

Lemma 7.2 (Annulus-to-tail excursion)**.** *There exist $K \geq 1$ and $z \geq 0$, depending on the fixed parameters but not on the tail threshold, such that for $t > 4K$,*

$$\inf_{2K\leq|y|\leq 4K} \mathbb{P}_y(\tau_{D_t} < \tau_C) \geq \frac{p_0}{2} e^{-\alpha z}\left(\frac{K}{t}\right)^{\alpha}, \quad C = [-K, K], \quad D_t = \{x : |x| > t\}. \tag{7.13}$$

The hitting times here start at time one.

Proof. Choose $z \geq 0$ so that (7.6) is at most $p_0/4$. Next choose $K \geq 1$ so that $\mathbb{Q}(V_\infty > K/2) \leq p_0/4$. Both choices precede the choice of t. Put $u = \log(t/K)$ and use the stopped event

$$F_u = \{\tau_u < \infty,\ S_j > 0\ (1 \leq j \leq \tau_u),\ V_{\tau_u} \leq K/2,\ O_u \leq z\}. \tag{7.14}$$

This event belongs to $\mathcal{F}_{\tau_u}$. The survival event in (7.5), minus the two exceptional events just bounded, is contained in F_u. The union bound thus gives $\mathbb{Q}(F_u) \geq p_0/2$.

For every initial point with $2K \leq |y| \leq 4K$, the pathwise inequality (7.2) yields on F_u

$$\begin{aligned} |X_j| &\geq \frac{3K}{2} e^{S_j} > K \qquad (1 \leq j \leq \tau_u), \\ |X_{\tau_u}| &> \frac{3K}{2}\frac{t}{K} > t. \end{aligned} \tag{7.15}$$

Thus F_u forces a visit to D_t before C. Since $S_{\tau_u} \leq u + z$ on F_u, the stopped identity (7.3) gives

$$\mathbb{P}(F_u) \geq e^{-\alpha(u+z)}\mathbb{Q}(F_u) \geq \frac{p_0}{2} e^{-\alpha z}(K/t)^{\alpha}. \tag{7.16}$$

This proves (7.13). The initial annulus lies outside both target sets. Under $\mathbb{P}$, τ_u need not be almost surely finite; (7.3) already restricts to the finite-crossing event. □

7.4 Stationary last-exit transfer

Lemma 7.3 (Last-exit inequality)**.** *For disjoint Borel sets C, D and $\tau_C^+ = \inf\{n \geq 1 : X_n \in C\}$, a stationary chain satisfies*

$$\pi(D) \geq \int_C \pi(\mathrm{d}x)\, \mathbb{P}_x(\tau_D < \tau_C^+). \tag{7.17}$$

Proof. Fix $N \geq 1$. Among trajectories with $X_N \in D$, classify those having a visit to C at a time $0, \ldots, N-1$ according to their last such visit. These events are disjoint. If the last visit occurs at $N-j$, stationarity and the Markov property express its probability as $\int_C \pi(\,\mathrm{d}x)\mathbb{P}_x(X_j \in D,\ j < \tau_C^+)$. Consequently

$$\begin{aligned}\pi(D) &\geq \int_C \pi(\,\mathrm{d}x) \sum_{j=1}^{N} \mathbb{P}_x(X_j \in D,\ j < \tau_C^+) \\ &\xrightarrow[N\to\infty]{} \int_C \pi(\,\mathrm{d}x)\mathbb{E}_x\left[\sum_{j\geq 1} \mathbf{1}_{\{X_j\in D,\ j<\tau_C^+\}}\right] \\ &\geq \int_C \pi(\,\mathrm{d}x)\mathbb{P}_x(\tau_D < \tau_C^+). \end{aligned} \tag{7.18}$$

The passage to the infinite sum uses monotone convergence. The stationary integral is unnormalized; no return-time or regeneration factor is omitted. □

Proposition 7.4 (Strictly positive tail constant)**.** *For the constant from proposition 6.2, $C_{\eta,\theta} > 0$.*

Proof. Take K, z from lemma 7.2, $C = [-K, K]$, and $J = \{y : 2K \leq |y| \leq 4K\}$. Lemma 5.1, with $R = K$ and $T = 4K$, gives

$$\inf_{x\in C} P_\theta(x, J) \geq d_K := 4K\varepsilon_{K,4K} > 0. \tag{7.19}$$

Here $4K$ is the Lebesgue measure of the two-component annulus J. The Markov property after the first step and (7.13) yield, uniformly for $x \in C$,

$$\mathbb{P}_x(\tau_{D_t} < \tau_C^+) \geq d_K \frac{p_0}{2} e^{-\alpha z} K^\alpha t^{-\alpha}, \qquad t > 4K. \tag{7.20}$$

Full support implies $\pi(C) > 0$. Applying lemma 7.3 now gives

$$\pi(D_t) \geq c_{\eta,\theta} t^{-\alpha}, \qquad c_{\eta,\theta} := \pi(C) d_K \frac{p_0}{2} e^{-\alpha z} K^\alpha > 0, \quad t > 4K. \tag{7.21}$$

Thus $\liminf_{t\to\infty} t^\alpha \pi(D_t) \geq c_{\eta,\theta} > 0$. Combining this with the existing finite limit in proposition 6.2 proves $C_{\eta,\theta} \geq c_{\eta,\theta} > 0$. □

8 Exact moment threshold and estimation error

For a nonnegative random variable R and $p > 0$, Tonelli's theorem gives

$$\mathbb{E}R^p = p \int_0^\infty t^{p-1}\mathbb{P}(R > t)\,\mathrm{d}t. \tag{8.1}$$

The positive limit in (1.7) makes this integral diverge for every $p \geq \alpha$, including the logarithmic divergence at $p = \alpha$. Proposition 4.1 proves finiteness for $0 < p < \alpha$.

A bounded translation preserves the exact tail constant. For $t > |\theta|$, the triangle inequality gives

$$\mathbb{P}(|X| > t + |\theta|) \leq \mathbb{P}(|X - \theta| > t) \leq \mathbb{P}(|X| > t - |\theta|). \tag{8.2}$$

Multiply by t^α and use $t^\alpha/(t \pm |\theta|)^\alpha \to 1$. Both endpoints converge to $C_{\eta,\theta}$, so the estimation-error tail has the same limit. The same tail integral, together with bounded-shift moment inequalities, gives its identical moment threshold. Since $\alpha < 2$, both second moments are infinite. This completes the proof of theorem 1.1.

9 The concrete nonzero-signal example

At $\theta = 1$, the labels are $Y = A + \delta \tanh A + E$, with

$$\mathbb{E}[AY] = 1 + \delta \mathbb{E}[A \tanh A] > 1. \tag{9.1}$$

They contain a nonconstant input-dependent regression signal. More generally, the population risk has the unique minimizer θ:

$$\begin{aligned} F_\theta(x) := \mathbb{E} f_\theta(x; A, E) &= \frac{1}{2} + \frac{1}{2}\mathbb{E}[\sigma(Ax) - \sigma(A\theta)]^2 \\ &\geq \frac{1}{2} + \frac{1}{2}(x - \theta)^2. \end{aligned} \tag{9.2}$$

Here independence and the mean-zero noise remove the cross term, and $\sigma' \geq 1$ together with $\mathbb{E}A^2 = 1$ proves the inequality. At $x = \theta$, the risk is exactly $1/2$, proving uniqueness.

At $\eta = 1$, the arithmetic in (3.5) sharpens to

$$\mathbb{E}\,\mathrm{Lip}(\Psi_\theta) < \frac{97}{100} + \frac{10007}{10^6} = 0.980007, \quad h_1(1) = \frac{4}{\sqrt{2\pi e}} < 0.97, \quad h_1(2) = 2. \tag{9.3}$$

In particular, $\theta = \eta = 1$ is a genuine nonzero-teacher example with an exact positive stationary estimation-error tail of index $\alpha(1) \in (1, 2)$, uniquely specified by $\mathbb{E}|1 - A^2|^{\alpha(1)} = 1$.

10 The precise symmetry statement

Oddness of σ and evenness of σ' imply the exact identity

$$\Psi_{-\theta, A, -E}(-x) = -\Psi_{\theta, A, E}(x). \tag{10.1}$$

The additive noise is symmetric and the invariant law is unique, so $\pi_{\eta,-\theta}$ is the reflection of $\pi_{\eta,\theta}$. At $\theta = 0$, one law is therefore symmetric, and for every $t > 0$, $\pi_{\eta,0}((t, \infty)) = \pi_{\eta,0}((-\infty, -t)) = \pi_{\eta,0}(|x| > t)/2$. Each one-sided tail constant is $C_{\eta,0}/2$. Reflection also gives $C_{\eta,-\theta} = C_{\eta,\theta}$ for the absolute tails.

At fixed $\theta \neq 0$, reflection changes the parameter and is not a symmetry of that model. In fact exact stationary symmetry about zero is impossible. Stationarity, the finite first moment, and averaging out the fresh mean-zero noise give

$$0 = \mathbb{E}\left[A\sigma'(AX)(\sigma(AX) - \sigma(A\theta))\right]. \tag{10.2}$$

Both terms are integrable because $\sigma' \leq 1 + \delta$, $|\sigma(v)| \leq (1 + \delta)|v|$, and $\mathbb{E}_\pi|X| < \infty$. If π were symmetric, $\mathbb{E}[A\sigma'(AX)\sigma(AX)] = 0$ since its integrand is odd in X for each fixed A. But $A\sigma'(AX)\sigma(A\theta)$ has the strict sign of θ almost surely when $A \neq 0$, contradicting (10.2). Equality of asymptotic one-sided constants would require a separate signed-tail argument; it is not deduced here.

11 Finite-time moment asymptotics

Write $h(p) = \mathbb{E}|M|^p$, $M = 1 - \eta A^2$, and $m_\alpha = \mathbb{E}[|M|^\alpha \log |M|]$. Let $X_0 = x_0 \in \mathbb{R}$ be deterministic. All finite-time moments exist by the uniform remainder bound and the Gaussian envelopes in Section 2.

Theorem 11.1 (Critical and supercritical moments)**.** *For every fixed (η, θ) in the parameter rectangle of Theorem 1.1,*

$$\lim_{n\to\infty} \frac{\mathbb{E}|X_n|^\alpha}{n} = \alpha m_\alpha C_{\eta,\theta} > 0.$$

For each $p > \alpha$, there is a constant $K_p(x_0) \in (0, \infty)$ such that

$$\lim_{n\to\infty} h(p)^{-n}\mathbb{E}|X_n|^p = K_p(x_0).$$

The same two limits, with the same constants, hold when $|X_n|$ is replaced by the estimation error $|X_n - \theta|$.

Proof. Define

$$g(x) = \mathbb{E}\big[|\Psi_\theta(x)|^\alpha - |Mx|^\alpha\big].$$

The coupled difference estimate (6.4) proves $|g(x)| \le K(1 + |x|^{\alpha-1})$. Dominated convergence on every compact set proves continuity. For $0 < r < \alpha$, the bound $|\Psi_\theta(x)| \le |M||x| + B_\theta$, together with $h(r) < 1$, gives $\mathbb{E}[|X_{n+1}|^r \mid X_n = x] \le \rho_r|x|^r + K_r$ for some $\rho_r < 1$. For $r \le 1$ use subadditivity; for $r > 1$ use $(u + v)^r \le (1 + \epsilon)u^r + C_{r,\epsilon}v^r$ with $(1 + \epsilon)h(r) < 1$. Thus $\sup_n \mathbb{E}|X_n|^r < \infty$. Choose $\alpha - 1 < r < \alpha$. This moment bound makes $g(X_n)$ uniformly integrable. Weak convergence $X_n \Rightarrow \pi$, established in Section 4, now yields

$$\mathbb{E}g(X_n) \longrightarrow \mathbb{E}_\pi g(X) = \alpha m_\alpha C_{\eta,\theta}.$$

Here the final equality is the integrable coupled-difference formula (1.8). Since $h(\alpha) = 1$, independence of the fresh map gives

$$\mathbb{E}|X_{n+1}|^\alpha - \mathbb{E}|X_n|^\alpha = \mathbb{E}g(X_n).$$

Telescoping and Cesàro averaging prove the critical limit.

For $p > \alpha > 1$, set $u_n = \|X_n\|_p$, $a = h(p)^{1/p} > 1$ and $b = \|B_\theta\|_p < \infty$. The triangle and reverse triangle inequalities give

$$|u_{n+1} - au_n| \le b.$$

Consequently $a^{-n}u_n$ is Cauchy and converges to some $\ell \ge 0$, with $|u_n - \ell a^n| \le b/(a - 1)$. The limit cannot be zero: $X_n \Rightarrow \pi$ and $\mathbb{E}_\pi|X|^p = \infty$ imply $\liminf_n \mathbb{E}|X_n|^p = \infty$, by applying weak convergence to $\min(|x|^p, R)$ and then sending $R \to \infty$. If $\ell = 0$, the displayed error bound would keep u_n bounded. Thus $\ell > 0$ and $K_p(x_0) = \ell^p \in (0, \infty)$.

Finally, at the critical exponent the difference $\big||x-\theta|^\alpha - |x|^\alpha\big|$ is bounded by $K_\theta(1+|x|^{\alpha-1})$, whose expectations are uniformly bounded. At $p > \alpha$, the L^p norms of X_n and $X_n - \theta$ differ by at most $|\theta|$. These observations transfer both normalizations to the estimation error. □

The critical-moment formula has an earlier affine antecedent in Buraczewski, Damek and Zienkiewicz [4], who also treat a class of Lipschitz iterated systems. Here we give a direct proof for the stated nonlinear SGD model; we make no exclusive-priority claim.

The critical linear asymptotic is sharper for this model than the $O(n^\alpha)$ upper bound stated for $\alpha > 1$ in [2, Proposition 10]. Normalized supercritical moments have affine antecedents; for example, the perpetuity maximum in [3, equation (3.33) and its proof] has a positive finite normalized-moment limit. That reflected affine recursion is not the signed nonlinear update studied here. We make no exclusive-priority claim for these elementary bounded-perturbation refinements.

12 Attribution and scope

This manuscript consolidates earlier author-directed Math Brain proofs for scalar asymptotically linear SGD. The forward tilted-walk, overshoot, and stationary last-exit mechanism in Section 7 was supplied by an earlier project Claude agent; the nonzero-teacher continuation rechecked the label, remainder, complete-step change of measure, and Gaussian minorization. The public source archive records the contribution history and provides the complete consolidated proof. Consolidation does not reassign discovery credit or count an earlier project solution a second time.

The external exact-tail input is the $N = 1$ specialization of [1, Theorem 3.1]; the model-specific verification and positive-constant proof are supplied here. The result is confined to the scalar model and parameter rectangle in (1.1). It does not vary the batch size or prove a theorem for arbitrary activations, steps, or network architectures.

Public repository and source availability

The complete manuscript and editable source are archived in the public Math Brain repository [5]. The paper-specific archive and PDF are given in [6]. The release includes exact arithmetic checks and a dated verification record.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The mathematical arguments, verification, literature comparisons, and writing were produced by AI systems in the author-directed Math Brain project, including the earlier Claude work credited above and OpenAI systems used for the continuation and this manuscript. The author selected the problem and commissioned and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission or publication rests with the author. The checks are internal research verification, not external peer review or proof-assistant formalization.

A Exact arithmetic

The following identities are rational arithmetic checks, not numerical approximations used in place of proof.

Quantity	Exact value
Conservative mixed-label bound $3 + 2\delta$	$1501/500 = 3.002$
Expected derivative perturbation $4\delta + 3\delta^2 + 2\delta(3 + 2\delta)$	$10007/10^6 = 0.010007$
Uniform contraction upper bound q	$99010707/10^8 = 0.99010707$
Contraction margin $1 - q$	$989293/10^8 = 0.00989293$
Upper bound at $\eta = 1$	$980007/10^6 = 0.980007$
Minimum of $h_\eta(2)$ on the interval	$19603/10000 = 1.9603$
Squared rational normal-bound margin in (3.2)	$7532467/125000000 > 0$

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